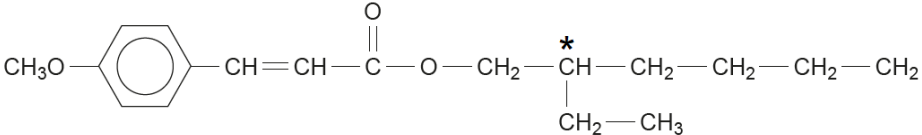


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	(a)	(i)		compound A does not absorb effectively in the UVA region			1	1		
		(ii)		C ₁₀ H ₁₀ O ₃		1		1		
		(iii)	I	structural isomerism is concerned with the position of atoms within a molecule (1) stereoisomerism is concerned with the positions that the atoms take up in space (1) neutral answers – reference to mirror images, chiral centres, E/Z	2			2		
			II		1			1		
			III	it does not rotate the plane of plane polarised light		1		1		
			IV	 <i>E</i> - form <i>Z</i> - form correct formulae (1) labelling (1)	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		electrophilic addition	1			1		
		(ii)		3 mol of I ₂ needed for 1 mol of the unsaturated oil moles of unsaturated oil = $\frac{0.0128}{3} = 0.00427$ (1) mass of unsaturated oil = 0.00427 × 885 = 3.78 g percentage of unsaturated oil = $\frac{3.78}{8.41} \times 100 = 45\%$ (1)		2		2	1	
		(iii)		3 NaOH (1) 3 C ₁₇ H ₃₃ COONa (1)		1	1	2		
		(iv)		propane-1,2,3-triol		1		1		
				Question 12 total	6	6	2	14	1	0